Frontier



GM Systems

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Stanford University
GM Systems, LLC

Organization location (we welcome applicants from anywhere in the world)

Stanford University: Stanford, CA
GM Systems: Aptos, CA

Name(s) of primary point(s) of contact for this application

Stanford: Rob Dunbar, co-PI
GM Systems: Gene Massion, corresponding co-PI

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Stanford: Research University
GM Systems: Ocean Engineering technology development company

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

The Development and Demonstration of an Open-Source Ocean Observing System for mCDR MRV Applications

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

We are developing an open-source ocean observing system optimized for mCDR applications (mOBS). mOBS consists of 3 major elements. First, an array of a new class of observing platform optimized for mCDR applications; second, an onshore data processing and modeling system that provides a range of data management and analysis capabilities. The third element is the foundation of tools, techniques, processes, documentation, and expertise required to successfully operate mOBS.

The overarching goals of mOBS:

1. Provide cost-effective, scalable, in-water observing platforms, sensors, an onshore data system, and near-field modeling capabilities that quantify the efficacy of mCDR technologies and, equally important, quantify the consequences of these mCDR technologies on local ecosystems.
2. Provide a cost-effective in-ocean test bed for the development of mCDR-relevant new, low-cost sensors, and MRV measurement protocols.
3. Create a mOBS open-source ecosystem (mOSE) to stimulate mCDR-specific technology development and accelerate impact at scale.
4. Create a non-profit commercial entity that leverages this OSE to offer a range of observing services.

mOBS is the result of conversations with a range of mCDR stakeholders, including mCDR OAE startups, and over 100 interviews with research, commercial, government, and NGO stakeholders in the mCDR space. These conversations made it clear that the absence of relevant and cost-effective platforms, sensors, and observing systems providing data for comprehensive MRV purposes is a primary bottleneck to scaling, funding, and building trust in mCDR technologies.

An additional impediment to advancing marine CDR/OAE is the lack of knowledge about alkalinity dispersion, mixing, and reactivity within the near-field mixing zone adjacent to injection sites. Isometric protocols do not require an understanding of this region, and there remains deep uncertainty in the efficiency of alkalinity transfer from injection site to the far-field ocean. The near-field mixing zone is where non-conservative behavior occurs, e.g., reactions, dissolution, precipitation, adsorption, and settling, and is also where environmental impacts on marine life will be greatest. mOBS is designed to directly monitor the near-field mixing zone, including biological impacts. Plume mixing and transport physics in the coastal zone are complex yet must be understood to convert moored profiler observations into 3D fields and fluxes of alkalinity. Such an understanding can also be used to guide alkalinity injection protocols to maximize efficiency. Our proposal includes the development of a Lagrangian particle tracking model of near-field alkalinity injection that will be

used to evaluate different mOBS deployment and injection scenarios.

Two critical capability gaps came out of our stakeholder conversations. First is the lack of an observing system optimized for use at mCDR OAE injection sites. Existing OAE pilot projects and scaled-up OAE plants being planned are mostly in shallow-water harbors and coastal environments with significant vessel traffic. Existing ocean observing platforms don't meet stakeholder needs in coastal locations. This includes moorings, wire walkers, bottom-mounted winches, gliders, AUVs, ROVs, Lagrangian profiling floats, and research vessels. A new class of observing platform optimized for mCDR applications is required. Development of this platform, described in more detail later in this application, is the key innovation of our mOBS project and will provide a step-change improvement in observing capabilities for the mCDR community.

The second capability gap is the lack of a complete observing system that provides an integrated solution optimized for mCDR applications, including data-assimilating modeling of the injection plume. mOBS integrates the new observing platform with an onshore data and modeling system to provide our stakeholders with defensible data fit for their purposes.

To date, observing systems for mCDR MRV applications have been implemented in a range of independent bespoke projects by several commercial mCDR startups and a few research groups. These systems meet a very specific set of narrow requirements using the technology and budgets available. A critical element of our approach is standing up an independent science and technology advisory team. The charter for this team is to make sure we are leveraging lessons learned, not repeating mistakes, and ensuring mOBS meets the needs of a broad range of mCDR applications.

mOBS addresses these capability gaps and directly advances Frontier's priority innovation area of "Improving the MRV toolkit for open system pathways" and supports the "Strategies to maximize alkalinity addition rate for OAC and IWAE". While our focus is on OAE technologies, mOBS can support other mCDR technologies, including marine ERW, freshwater AE, and direct ocean capture. It can also support blue-C accounting.

We have a work breakdown structure for all elements of this project that includes the following high-level tasks.The first iterations of task 1 and 2 are complete and we are currently part way through task 3.

1. Identify our stakeholders and develop an understanding of their needs.
2. Develop concepts that may meet our stakeholder needs. Iterate those concepts with our stakeholders until a preferred design emerges.
3. Finish and review the preliminary design for mOBS. A critical part of this step is identifying risks and implementing a risk mitigation plan.
4. Implement injection plume mixing and dispersion/reactivity model and explore deployment scenarios.
5. Build, test, improve, and repeat until we can verify the final design meets our stakeholder needs. This applies to the profiling float as well as the co-developing model.
6. Critical design review
7. Initial at-sea testing of mOBS
8. Demonstrate mOBS in an operational environment. We have started conversations with 2 mCDR OAE startups to co-design a demonstration at one of their mCDR OAE pilot plants. All data and the field demonstration report will be publicly available on an early version of the mOBS OSE website. In addition, the engineering documentation, mechanical, electrical, and software, will be available on this website. This would bring mOBS to technical readiness level 7 (TRL 7).

9. Finish building the mOBS (OSE)
10. Develop a commercial entity that can leverage this OSE to provide a range of observing services to the mCDR community (TRL 9).

The field demonstration, task 8, is the critical step to unlock step-change advancements in cost and scale in the mCDR observing space. We are seeking Frontier funds to complete tasks 3-8. Completing these tasks will enable future funding to complete tasks 9 and 10.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

The most significant differentiator in this project is the development of the new class of observing platform identified earlier. The requirements for the mOBS observing platform came out of many conversations with our stakeholders and colleagues in the mCDR space and include the following.

1. Support CTD, oxygen, pCO_2, fluorescence, backscatter, turbidity, pH, dissolved inorganic carbon, total alkalinity, and nutrient sensors. Along with 1 or 2 additional sensors to enable sensor-to-sensor intercomparisons and new sensor evaluation.
2. Allow for continuous profiling at a user specified vertical velocity.
3. Provide a depth hold capability at a user specified depth for a user specified time for sensors with a long equilibration time e.g. pCO_2.
4. Able to work in shallow, high-biofouling coastal and harbor environments at less than 5 meters to 100 meters of water depth.
5. Provide 180 full water depth (seabed to sea surface) profiles defined by the user over a period of up to 1 year or until biofouling prevents effective operation. The profiling schedule shall be defined by the user, e.g., 1 profile every hour for 180 hours or 1 profile every other day for 1 year.
6. Telemeter the complete set of science and engineering data immediately after each profile.
7. Maintain a spatially dense array of full water column profilers in the mixing zone surrounding an OAE plant pipe outfall including profilers in the transition zone to the far-field.
8. Provide a continuous set of measurements from the entire array over timespans of months to years.
9. Minimize overall life cycle costs. Particularly deployment, recovery, and refurbishment costs. Autonomous platforms are preferred over platforms that require 24/7 pilots, e.g., gliders and AUVs.

The observing platform we are developing as part of this work meets all the requirements above. Unique capabilities of this platform include the following.

1. The ability to make measurements from and slightly below the seabed, through the water column up to and slightly above the sea surface in water depths from less than 5 to 100 meters.
2. Support the broad suite of sensors required for mCDR observations. Including physically

large sensors like the current generation of nutrient and pCO_2 sensors. Additional sensor ports are available for new sensor development and inter-sensor comparisons.

3. This new platform is relatively small, roughly the size of a large microwave oven and weighs less than 40 kg. It can be deployed from a small boat (< 10 meters) equipped with a small davit. Refurbishment prior to redeployment consists of recharging the batteries, biofouling removal, recalibrating sensors, and replacing parts if required.

The unique capabilities of the mOBS platform particularly relevant to mCDR applications include the following.

1. The number and size of sensors required to support mCDR observations is quite large. The mOBS platform sensor payload capability is best-in-class compared to other profiling platforms we are aware of that can work in the mCDR relevant environment we are targeting.
2. At a recent Carbon to Sea webinar, Vesta described the importance of making measurements in the pore water below up to <1 cm above the seafloor. The mOBS platform can park directly on the seafloor and could support a new instrument with this sub-bottom profiling capability for pH and other relevant parameters. We believe this capability combined with the other capabilities of the mOBS platform is not only best-in-class, but unique. We would be excited to team with a group developing such an instrument.
3. The current generation of pCO_2 sensors require 30 seconds to several minutes to make a measurement. The observing platform has to maintain a reasonably constant 3D position during this measurement period. This capability combined with the large sensor suite and profiling capability of the mOBS platform combined with the ability to work in crowded, busy harbors or coastal areas is, to the best of our knowledge, unique.
4. Current quality control processes for oxygen and optical radiation sensors generally require in-air measurements in air. The ability to mount these sensors 10s of centimeters above the mOBS platform's waterline enables this capability. In addition, this capability enables the deployment of sensors that measure the flux of relevant parameters across the sea-surface/air boundary, e.g., the ProOceanus CO2-Pro ATM Again, we believe this capability combined with the large sensor suite and profiling capability of the mOBS platform combined with the ability to work in crowded, busy harbors or coastal areas is unique.

As capable as this new observing platform is, we recognize that the final product our stakeholders need is defensible data and data products fit for their purposes. The mOBS onshore data system is based on Ocean Data View (ODV). ODV is an existing open-source data system optimized for oceanographic data and widely used across the ocean sciences community. ODV already has extensive data management, exploration, visualization, quality control, and export capabilities. We will be enhancing these built-in capabilities by providing several mCDR-specific data products. These include integrating existing community-developed quality control (QC) procedures, like those developed by the Argo and BGC-Argo communities, into the existing ODV QC functionality. A second enhancement is the ability to export mOBS data and data products in a format compatible with the models being used and developed by the mCDR community.

The project's postdoctoral researcher will determine which plume modeling code is most useful for our specific goals (OAE by coastal point source injection) and implement a working model at Stanford. We do not begin from scratch. Moniz et al. (2014) developed analytical shear-flow dispersion and particle-tracking models to forecast lateral dispersion of a Stanford-led dye tracer

injection experiment in Monterey Bay, one of our proposed test sites. Additionally, Steinbuck et al. (2011) led a Stanford project that developed a Lagrangian particle tracking model to explore vertical diffusion and vertical shear from internal waves at time and space scales appropriate for our mOBS system goals. These codes are available for our use. Co-authors Monismith, Koseff, and Fong are long-term ocean biogeochemistry/hydrodynamics research collaborators with Dunbar and will engage and assist in this modeling effort as part of our overall interest in mCDR. For alkalinity injection involving buoyant plumes (warm or low salinity outfalls), we refer to Liang et al. (2018), who used 3D turbulence-resolving Lagrangian particle tracking and dispersion models in bounded shear currents to examine dispersion in the ocean surface boundary layer.

The post-doc will also explore the use of Oceananigans (Wagner et al., 2025) for our purposes. Oceananigans is an easily adapted numerical ocean model developed for community use by the Climate Modeling Alliance. Oceananigans can efficiently simulate all scales of ocean motion, ranging from millimeter-scale turbulence in a small box to planetary-scale ocean circulation.

Once a modeling framework has been selected, the postdoc will implement a model of mixing and transport within a domain adjacent to a point source injection site. The model will be capable of simulating plume physics under a range of ocean conditions as well as transport under several different injection scenarios, e.g., variable water depth, feedstock type, warm outfalls, desalination outfalls, and wastewater outfalls. Particle tracking experiments will inform the design of OAE outfall MMRV, e.g., the use of mOBS (how many moored profiling floats and where to deploy them) and AUVs (where to deploy and how often), as well as how to intake mOBS data into a data processing and QC regimen. Eventually the model will assimilate field data from mOBS and can be used to alter observation protocols in real-time as needed and to produce quantitative estimates of alkalinity fluxes to the far-field ocean as well as their uncertainty. Reactivity models of this nature do not yet exist. Once a particle tracking and plume mixing model is running, Dunbar and alkalinity feedstock research partner Matt Kanan will advise on how to implement kinetic and equilibrium parameterizations for several precipitation/dissolution systems known to be important for alkalinity feedstock materials. This reactivity work will be a preliminary step and be attempted if time is available.

We see the co-development of a fit-to-purpose plume transport, dispersion, and reactivity model as tightly coupled to the development of the full mOBS system, as it can be used to inform the design and operational features of the profiling floats and their placement for MMRV, as well as to assimilate and process data incoming from mOBS to quantify alkalinity fluxes.

A fully developed and user-tested near-field alkalinity injection, transport, mixing, and reactivity model is beyond the time frame and resources of this RFP. We propose to start the work on model development with a skilled post-doc as we co-design and build out the full mOBS system.

To the best of our knowledge and that of our colleagues, no ocean observing system that combines this level of capability for mCDR applications is available or in development either commercially or in the oceanographic research community. In fact, we believe mOBS and the WPF in particular will provide a step-change improvement in ocean observing that can scale to support a broad range of oceanographic applications, including coastal carbon cycling, fishery management, and harmful algal bloom, hypoxia, and marine heat wave studies.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Developing the new mOBS observing platform is the biggest risk in our project. The mCDR stakeholder requirements are challenging. Fortunately, the new class of observing platform we are developing for mOBS integrates 2 technologies we have previously developed. Our mitigation plan for the mOBS platform builds on the lessons we learned from both those projects. We have completed initial structural, hydrodynamic, and motion analysis. The next step is to simulate the platform motion over a range of sea states from near calm to past our goal of full operations in WMO sea state 4 using ProteusDS. ProteusDS is a state-of-the-art hydrodynamic simulator that analyzes buoy, mooring, and vessel motion in user-specified wind and wave conditions. Those simulation results will drive the platform's hydrodynamic and control system preliminary design to ensure stable behavior. The next mitigation step is testing on the ocean motion simulator (OMS) we have designed. The OMS allows the platform control systems to be tested, optimized, and validated against the ProteusDS simulations. At that point, we will be ready for at-sea verification.

The other major elements of mOBS, the data QC, visualization, and export system, as well as the integrated near-field modeling system, will require a development and test plan to meet stakeholder needs. We have a strong environmental fluid mechanics research group at Stanford Oceans and Engineering and substantial experience with developing and enhancing flow models of all types. Stanford has multiple data science centers (as well as open science support units) that we have consulted with regarding automation for data transmission, QC, visualization, as well as value-added products like estimations of alkalinity transfer to the ocean, along with uncertainty quantification. Finding the right postdoc is a risk at this time when the best young scientists are fleeing the country due to the current "war on science." Nevertheless, we have some candidates in mind and believe that the proposed research will be attractive to hydrodynamic modelers wishing to help with the mitigation of climate change.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	The primary deliverable is a report documenting the successful completion of the field demonstration at an OAE pilot project as described above. We will also deliver a model of near-field plume mixing and fate optimized for use with a mOBS mooring array.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	The bottleneck mOBS is addressing is the lack of a comprehensive system integrating platforms, sensors, data systems, and models into a cost-effective, scalable ocean observing system for mCDR applications. mOBS will be the first

	system that resolves this bottleneck.]
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	1. The Frontier project will be complete after the field demonstration is complete and the data, model, and final report are published on the mOBS website 21 months after the award of contract. Intermediate milestones include the following. 2. Completion of the preliminary design review, 4 months after award 3. Completion of the WPF lab testing. 9 months after award. 4. Completion of the critical design review, 12 months after award 5. Completion of the initial at-sea testing. 19 months after award of contract.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$750k